

HU Collection

TMUA Mock Set J Paper 1

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	E	B	F	A	D	C	F	B	E

Q	11	12	13	14	15	16	17	18	19	20
Ans	A	D	B	C	F	E	A	D	C	B

Each solution gives the intended route and names the trap the question was built around. Every question is pitched at the hard end of the TMUA, and they are loosely ordered by increasing difficulty.

Question 1. Answer C.

Factorise the second cubic first: $x = 1$ is a root, and $x^3 - 2x^2 - 5x + 6 = (x - 1)(x - 3)(x + 2)$. So the three pairwise sums are 1, 3 and -2 in some order. Adding them gives $2(\alpha + \beta + \gamma) = 2$, so $\alpha + \beta + \gamma = 1$, and each root is 1 minus the sum of the other two: the roots are 0, -2 and 3. Hence $x^3 + ax^2 + bx + c = x(x + 2)(x - 3)$, and $1 + a + b + c$ is its value at $x = 1$, namely $1 \cdot 3 \cdot (-2) = -6$. Option D is what you get by evaluating the *second* cubic at 1, or by taking $\alpha + \beta + \gamma$ to be the sum of the new roots rather than half of it; option B forgets the leading 1.

Question 2. Answer E.

Let θ be the angle AOB in radians, with $0 < \theta < \pi$. The arc has length $r\theta$ and the chord has length $2r \sin \frac{\theta}{2}$, so we need $\theta + 2 \sin \frac{\theta}{2} = \frac{\pi}{3} + 1$. Both terms on the left increase with θ on this range, so there is only one solution, and $\theta = \frac{\pi}{3}$ visibly works since $\sin \frac{\pi}{6} = \frac{1}{2}$. The triangle has area $\frac{1}{2}r^2 \sin \theta = \frac{\sqrt{3}}{4}r^2$ and the sector has area $\frac{1}{2}r^2\theta = \frac{\pi}{6}r^2$, so the segment has area $\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)r^2$.

The ratio is $\frac{\pi/6 - \sqrt{3}/4}{\sqrt{3}/4} = \frac{2\pi}{3\sqrt{3}} - 1 = \frac{2\sqrt{3}\pi - 9}{9} \approx 0.21$. Guessing θ from the numbers is fine, but the monotonicity is what turns the guess into the answer. Option B compares the segment with the sector instead of the triangle.

Question 3. Answer B.

The distance from the x -axis is $|y|$, so $x^2 + (y - 2)^2 = (|y| + 1)^2$. If $y \geq 0$ this gives $x^2 - 4y + 4 = 2y + 1$, that is $y = \frac{x^2 + 3}{6}$, which is indeed non-negative. If $y < 0$ it gives $x^2 = 2y - 3$, which would need $y \geq \frac{3}{2}$, a contradiction; so there is no second branch, and the path is the single parabola $y = \frac{x^2 + 3}{6}$. It meets $y = 2$ at $x = \pm 3$, and the area is

$$\int_{-3}^3 \left(2 - \frac{x^2 + 3}{6}\right) dx = \frac{1}{6} \int_{-3}^3 (9 - x^2) dx = \frac{1}{6} \cdot 36 = 6.$$

The modulus looks as though it should produce two pieces of curve; checking that the lower piece is empty is the step to take care over.

Question 4. Answer F.

Let a be the probability that Amy eventually wins when it is Amy's turn, and b the same probability when it is Ben's turn. On her turn Amy wins with probability $\frac{1}{6}$, rolls again with probability $\frac{1}{6}$, and hands over with probability $\frac{4}{6}$, so $a = \frac{1}{6} + \frac{1}{6}a + \frac{2}{3}b$. On Ben's turn Amy loses with probability $\frac{2}{6}$, Ben rolls again with probability $\frac{1}{6}$, and the die returns to Amy with probability $\frac{3}{6}$, so $b = \frac{1}{6}b + \frac{1}{2}a$, giving $b = \frac{3}{5}a$. Then $\frac{5}{6}a = \frac{1}{6} + \frac{2}{5}a$, so $\frac{13}{30}a = \frac{1}{6}$ and $a = \frac{5}{13}$. Without the extra rule the same method gives $\frac{3}{8}$, option B; option A is the naive ratio of the single-roll chances $\frac{1}{6} : \frac{2}{6}$.

Question 5. Answer A.

The logarithm needs $x + 2 > 0$ and a base x^2 that is positive and not 1: so $x > -2$, $x \neq 0$ and $x \neq \pm 1$. The tempting step $\log_{x^2} y = \frac{1}{2} \log_x y$ is only available when $x > 0$, and using it throws away every negative solution; that is option B. Work with the base x^2 instead. Since $(x^2)^{1/2} = |x|$, the inequality says $x + 2 < |x|$ when the base exceeds 1, and $x + 2 > |x|$ when the base is less than 1. If $|x| > 1$: for $x > 1$ it reads $x + 2 < x$, impossible, while for $-2 < x < -1$ it reads $x + 2 < -x$, which holds throughout. If $0 < |x| < 1$: $x + 2 > |x|$ holds for every such x . So the solution set is $-2 < x < -1$ together with $-1 < x < 0$ and $0 < x < 1$.

Question 6. Answer D.

$\frac{dy}{dx} = 4x^3 - 12ax^2 + 12x = 4x(x^2 - 3ax + 3)$. Near $x = 0$ the bracket is close to $3 > 0$, so the derivative changes from negative to positive and the origin is always a local minimum. A stationary point that is not a turning point needs $\frac{dy}{dx}$ to vanish without changing sign, which here means the bracket has a repeated root: $9a^2 = 12$, so $a = \pm \frac{2}{\sqrt{3}}$, and the repeated root is $x = \frac{3a}{2} = \pm \sqrt{3}$. For $a = \frac{2}{\sqrt{3}}$, $y = 9 - 4 \cdot \frac{2}{\sqrt{3}} \cdot 3\sqrt{3} + 18 = 9 - 24 + 18 = 3$. For $a = -\frac{2}{\sqrt{3}}$ the curve is the reflection of this one in the y -axis, so the point is $(-\sqrt{3}, 3)$. Either way the answer is 3. Option A is the origin, which is stationary for every a but is always a minimum.

Question 7. Answer C.

Fix x and look at the vertical cross-section. The condition is $|2y - x^2 - 1| \leq 3 - |x|$, which needs $|x| \leq 3$ and then says that y lies within $\frac{3-|x|}{2}$ of $\frac{x^2+1}{2}$: an interval of length $3 - |x|$. The parabola only decides *where* each cross-section sits, not how long it is, so the area is

$$\int_{-3}^3 (3 - |x|) dx = 9.$$

Sliding vertical strips up or down never changes the area they cover. Option E forgets that the interval for y is half as long as the interval for $2y$.

Question 8. Answer F.

Write $a = BC$ and $b = CA$. The cosine rule at C gives $144 = a^2 + b^2 - \sqrt{2}ab$, a quadratic in b : $b^2 - \sqrt{2}ab + (a^2 - 144) = 0$, with discriminant $2a^2 - 4(a^2 - 144) = 576 - 2a^2$. Real triangles need $a \leq 12\sqrt{2} \approx 16.97$, so $a \leq 16$. The two roots have product $a^2 - 144$ and positive sum. If $a < 12$ the product is negative, so exactly one root is positive: one triangle. If $a = 12$ the roots are 0 and $12\sqrt{2}$, so again one triangle. If $12 < a \leq 16$ the product and the discriminant are both positive: two different triangles. The count is $12 + 2 \times 4 = 20$. This is the ambiguous case of the sine rule, counted: a second triangle exists exactly when $AB < BC < \frac{AB}{\sin 45^\circ}$. Option B counts one triangle for each possible a .

Question 9. Answer B.

A chord subtending 60° at the centre forms an equilateral triangle with the centre, and its distance from the centre is $R \cos 30^\circ$. That distance is 1, half the side of the square, so $R = \frac{2}{\sqrt{3}}$. The corners are at distance $\sqrt{2} > R$ from the centre, so the common region is bounded alternately by the four chords and by four arcs. Split it from the centre into four triangles and four sectors. Each triangle has two sides R and apex angle 60° , so area $\frac{1}{2}R^2 \sin 60^\circ = \frac{\sqrt{3}}{3}$. The four sectors share the remaining

$360^\circ - 4 \times 60^\circ = 120^\circ$, so together they have area $\frac{1}{2}R^2 \cdot \frac{2\pi}{3} = \frac{4\pi}{9}$. The total is $\frac{4\pi}{9} + \frac{4\sqrt{3}}{3} \approx 3.71$, a little less than the area 4 of the square. Option F exceeds 4 and cannot be right.

Question 10. Answer E.

Every sign change happens at a multiple of $\frac{\pi}{6}$: $\cos x$ changes sign at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$, $\sin 2x$ at $\frac{\pi}{2}$, π and $\frac{3\pi}{2}$, and $\cos 3x$ at $\frac{\pi}{6}$, $\frac{\pi}{2}$, $\frac{5\pi}{6}$, $\frac{7\pi}{6}$, $\frac{3\pi}{2}$ and $\frac{11\pi}{6}$. On the eight intervals between consecutive points the number of positive values is, in order, 3, 2, 1, 0, 1, 2, 1, 2. Exactly two are positive on $(\frac{\pi}{6}, \frac{\pi}{2})$, $(\frac{7\pi}{6}, \frac{3\pi}{2})$ and $(\frac{11\pi}{6}, 2\pi)$, of lengths $\frac{\pi}{3}$, $\frac{\pi}{3}$ and $\frac{\pi}{6}$. The total is $\frac{5\pi}{6}$. Option D counts where *at least* two are positive, which adds the opening interval $(0, \frac{\pi}{6})$.

Question 11. Answer A.

Year k 's planting, $P_k = 100(1.05)^{k-1}$, loses a tenth at the end of each year from year k to year n , so it contributes $(0.9)^{n-k+1}P_k$ to T_n . Hence, writing $j = n - k$ and using $\frac{P_k}{P_n} = (1.05)^{-j}$,

$$\frac{T_n}{P_n} = \sum_{k=1}^n (0.9)^{n-k+1} \frac{P_k}{P_n} = 0.9 \sum_{j=0}^{n-1} \left(\frac{0.9}{1.05} \right)^j.$$

This is a geometric series with ratio $\frac{0.9}{1.05} = \frac{6}{7}$, so the limit is $0.9 \cdot \frac{1}{1-6/7} = 6.3$. Option C forgets that each year's planting also suffers that year's deaths; option E ignores the growth in planting.

Question 12. Answer D.

Work with $P(x) = \int_0^x p(t) dt$ instead of p . It is a polynomial of degree 4 with $P(0) = 0$, and the three conditions say $P(1) = P(2) = P(3) = 0$. So $P(x) = kx(x-1)(x-2)(x-3)$ for some constant k , and $p = P'$. By the product rule, $P'(x)$ is k times the sum of the four products of three of the factors. At $x = 0$ only $(x-1)(x-2)(x-3)$ survives, giving $p(0) = -6k = 6$, so $k = -1$. At $x = 4$ the four products are $3 \cdot 2 \cdot 1$, $4 \cdot 2 \cdot 1$, $4 \cdot 3 \cdot 1$ and $4 \cdot 3 \cdot 2$, which sum to 50, so $p(4) = -50$. Option B is $P(4)$; option E drops the sign of k .

Question 13. Answer B.

Both cones have slant height 3. The arcs of the two sectors add up to the circumference 6π , and each arc becomes the circumference of a base, so the base radii satisfy $2\pi r_1 + 2\pi r_2 = 6\pi$, that is $r_1 + r_2 = 3$. The heights are $\sqrt{9 - r_1^2}$ and $\sqrt{9 - r_2^2}$. If the first cone is the taller, then $9 - r_1^2 = 4(9 - (3 - r_1)^2)$, which simplifies to $r_1^2 - 8r_1 + 3 = 0$, so $r_1 = 4 - \sqrt{13} \approx 0.39$ and $r_2 = \sqrt{13} - 1$. The volume ratio is

$$\frac{r_1^2 h_1}{r_2^2 h_2} = 2 \left(\frac{4 - \sqrt{13}}{\sqrt{13} - 1} \right)^2 = 2 \left(\frac{\sqrt{13} - 3}{4} \right)^2 = \frac{11 - 3\sqrt{13}}{4} \approx 0.046.$$

The taller cone is the narrow one, and it holds less than a twentieth of the other's volume. Option A takes the heights as the whole story; option E is the reciprocal.

Question 14. Answer C.

Let the sides enclosing the 120° angle be a and b , with $s = a + b$ and $p = ab$. The third side is $c = 12 - s$, and the cosine rule gives $c^2 = a^2 + b^2 + ab = s^2 - p$. So $(12 - s)^2 = s^2 - p$, that is $p = 24s - 144$. The area is $\frac{1}{2}ab \sin 120^\circ = \frac{\sqrt{3}}{4}p$, so we want p as large as possible. But real a and b with sum s need $p \leq \frac{s^2}{4}$, so $24s - 144 \leq \frac{s^2}{4}$, which means $s^2 - 96s + 576 \geq 0$, that is $s \leq 48 - 24\sqrt{3}$ (the other range makes c negative). Since p increases with s , the best is $s = 48 - 24\sqrt{3}$, where $a = b$ and $p = \frac{s^2}{4} = 144(2 - \sqrt{3})^2$. The area is $\frac{\sqrt{3}}{4} \cdot 144(7 - 4\sqrt{3}) = 252\sqrt{3} - 432 \approx 4.48$. Option A is the equilateral triangle, which is not allowed; option F is the value of ab .

Question 15. Answer F.

Both sides are even functions of x , and $x = 0$ is not a solution, so count the positive solutions and double. For $x > 0$ the right-hand side is $|\frac{x}{\pi} - 2|$: a V with its vertex at $(2\pi, 0)$, equal to 2 at $x = 0$

and $x = 4\pi$, and above 2 beyond 4π , where there are no solutions. The left-hand side is a row of arches of height 2, one on each interval $[k\pi, (k+1)\pi]$. On $[0, \pi]$ the V falls from 2 to 1 and crosses the first arch twice. On $[\pi, 2\pi]$ it falls from 1 to 0: one crossing as the arch rises, and then the graphs meet again at $x = 2\pi$, where both are 0. Just either side of 2π the arch is steeper than the V, so this is a single touching point, not a crossing. On $[2\pi, 3\pi]$ the V rises from 0 to 1 and crosses once as the arch comes down, and on $[3\pi, 4\pi]$, rising from 1 to 2, it crosses twice. That makes $2 + 2 + 1 + 2 = 7$ positive solutions and 14 in all. Option B misses the touching points at $\pm 2\pi$; option E counts each of them twice.

Question 16. Answer E.

For $|x| \leq 1$ the curve is $y = 2 + x^2$, and for $1 \leq |x| \leq 2$ it is $y = 4 - x^2$: an M shape with peaks of height 3 at $x = \pm 1$, a dip to 2 at $x = 0$, and feet at $x = \pm 2$. If the top of the rectangle is at height h , the whole top edge must stay under the curve. When $h > 2$ the region at height h splits into two pieces, one under each peak, each narrower than $\sqrt{2}$, so these rectangles have area less than $3\sqrt{2}$. When $h \leq 2$ the top edge can run from $-\sqrt{4-h}$ to $\sqrt{4-h}$, giving area $A = 2t(4-t^2)$ with $t = \sqrt{4-h} \geq \sqrt{2}$. Here $A'(t) = 8 - 6t^2 < 0$, so the best is $t = \sqrt{2}$: area $2\sqrt{2} \cdot 2 = 4\sqrt{2} \approx 5.66$. Maximising $2t(4-t^2)$ without the constraint gives $t = \frac{2}{\sqrt{3}}$ and $\frac{32\sqrt{3}}{9}$, option B, but that rectangle has height $\frac{8}{3}$ and pokes through the dip at $x = 0$.

Question 17. Answer A.

For $x \leq 3$ the modulus is $3 - t$ throughout the range of integration, and this holds for negative x too, so

$$F(x) = \int_0^x (-t^2 + 4t - 3) dt = -\frac{x^3}{3} + 2x^2 - 3x = -\frac{x}{3}(x-3)^2.$$

This is positive for $x < 0$, negative for $0 < x < 3$, and zero at 0 and 3. For $x > 3$, $F(x) = F(3) + \int_3^x (t-1)(t-3) dt$, and the integrand is positive there, so $F > 0$. The region therefore lies below the axis between 0 and 3, and its area is

$$\int_0^3 \frac{x}{3}(x-3)^2 dx = \frac{1}{3} \left[\frac{x^4}{4} - 2x^3 + \frac{9}{2}x^2 \right]_0^3 = \frac{1}{3} \cdot \frac{27}{4} = \frac{9}{4}.$$

The value $F(3) = 0$ is the surprise: the curve returns to the axis exactly at the zero of the modulus, crossing it there with a horizontal tangent. Option B is the depth of the region, $-F(1) = \frac{4}{3}$.

Question 18. Answer D.

Since $y \geq |x^2 - 1| \geq 0$, the first condition is just $y \leq 3 - |x|$, and by symmetry we can work with $x \geq 0$ and double. For $0 \leq x \leq 1$ the lower boundary is $1 - x^2$, which stays below $3 - x$, since the gap $x^2 - x + 2$ is always positive. For $x \geq 1$ it is $x^2 - 1$, which reaches $3 - x$ where $x^2 + x - 4 = 0$, at $c = \frac{\sqrt{17}-1}{2}$. So the area is

$$2 \left[\int_0^1 (2 - x + x^2) dx + \int_1^c (4 - x - x^2) dx \right].$$

The first integral is $\frac{11}{6}$. For the second, use $c^2 = 4 - c$ and $c^3 = 5c - 4$ to get $\left[4x - \frac{x^2}{2} - \frac{x^3}{3} \right]_1^c = \frac{17c}{6} - \frac{2}{3} - \frac{19}{6} = \frac{17\sqrt{17}-63}{12}$. The total is $2 \left(\frac{11}{6} + \frac{17\sqrt{17}-63}{12} \right) = \frac{17\sqrt{17}-41}{6} \approx 4.85$. Reducing powers of c with its own equation keeps the surds under control. Option E drops the modulus and uses $x^2 - 1$ all the way across.

Question 19. Answer C.

Substituting $x^2 = y + 2$ into the circle gives $y^2 + y - 2 = 0$, so $y = 1$ or $y = -2$. The value $y = -2$ gives only $x = 0$: the parabola touches the circle at its lowest point. The value $y = 1$ gives $x = \pm\sqrt{3}$.

Between these the parabola lies inside the circle and the region runs from the parabola up to the upper semicircle, while for $\sqrt{3} < |x| \leq 2$ the parabola is above the circle and there is nothing. So the area is

$$\int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{4-x^2} dx - \int_{-\sqrt{3}}^{\sqrt{3}} (x^2 - 2) dx.$$

The first integral is an area under an arc of the circle: the sector between the radii to $(\pm\sqrt{3}, 1)$, of angle $\frac{2\pi}{3}$ and area $\frac{4\pi}{3}$, plus two right-angled triangles with legs $\sqrt{3}$ and 1, of total area $\sqrt{3}$. The second integral is $2(\sqrt{3} - 2\sqrt{3}) = -2\sqrt{3}$. The area is $\frac{4\pi}{3} + \sqrt{3} + 2\sqrt{3} = \frac{4\pi}{3} + 3\sqrt{3}$. No antiderivative of $\sqrt{4-x^2}$ is needed; reading it as an area of the circle is the whole technique. Option A leaves out the parabola's contribution, and option E is the rest of the disc.

Question 20. Answer B.

Let $g(x) = |x^2 - 6|x| + 5| = |(|x| - 1)(|x| - 5)|$. Its graph is even: it touches the axis at $x = \pm 1$ and ± 5 , has arches of height 4 over $[1, 5]$ and $[-5, -1]$ with peaks at $x = \pm 3$, a peak of 5 at $x = 0$, and grows like x^2 outside $[-5, 5]$. Since g is above the line far out on both sides, a line of gradient 1 usually meets the graph an even number of times; an odd count needs a point where the two meet without crossing: a tangency to an arch, or a corner of the graph (the four zeros and the peak at the origin). Tangency to the right arch $y = -x^2 + 6x - 5$ needs $-2x + 6 = 1$, at $(\frac{5}{2}, \frac{15}{4})$, so $k = \frac{5}{4}$; tangency to the left arch happens at $(-\frac{7}{2}, \frac{15}{4})$, so $k = \frac{29}{4}$. The lines through $(1, 0)$, $(5, 0)$, $(-1, 0)$ and $(-5, 0)$ have $k = -1, -5, 1, 5$, and the line through $(0, 5)$ also has $k = 5$. Now count. $k = \frac{29}{4}$: the tangency plus one crossing on each side, 3. $k = \frac{5}{4}$: the tangency plus four crossings, 5. $k = -1$: $x = 1, 4$ and 6, so 3. $k = -5$: only $x = 5$. $k = 1$: five solutions. $k = 5$: four, namely $-5, -2, 0$ and 7. So exactly three solutions occur for $k = \frac{29}{4}$ and $k = -1$, with sum $\frac{25}{4}$. Option A misses the corner case $k = -1$.